

# Chi-Yao (Trey) Huang

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## SUMMARY

Ph.D. researcher in representation learning, embodied AI, and world models. My research develops interpretable and controllable latent-space learning for embodied AI, connecting spatial perception, semantic understanding, and reasoning through structured visual representations. Previously a Senior R&D Engineer at HTC VIVE, where I developed core VO/SLAM tracking systems deployed across major commercial VR/AR product lines.

## TECHNICAL SKILLS

**Research:** World models, representation learning, contrastive learning, generative models, foundation models  
**Computer Vision & Robotics:** SLAM, visual odometry, 3D reconstruction, sensor fusion, path planning, aerial robotics, humanoid robotics

**Programming:** Python, C/C++, JavaScript, MATLAB

**Tools & Frameworks:** PyTorch, OpenCV, Git, ROS, Ceres, Eigen, Docker, OpenGL, SIMD optimization, CEVA/Hexagon DSP

## EDUCATION

**Ph.D. in Computer Science** 2023 – PRESENT

Arizona State University, Tempe, AZ – Advisor: Prof. Yezhou Yang

**M.S. in Robotics and Autonomous Systems** 2021 – 2023

Arizona State University, Tempe, AZ

**M.S. in Mechanical Engineering** 2015 – 2017

National Taiwan University, Taipei, Taiwan

**B.S. in Mechanical Engineering** 2010 – 2014

National Sun Yat-Sen University, Kaohsiung, Taiwan

## SELECTED PUBLICATIONS

**Spatial World Model: Spatially Structured Latent Dynamics and Planning**

**Chi-Yao Huang** et al. *Under review at ICLR 2027*. Developed a reward-free world model whose latent space reflects the spatial structure of the physical environment, enabling interpretable dynamics and efficient goal-directed planning. ([Web](#) | [Paper](#) | [Code](#) | [Talk](#))

**Domain Expansion: A Latent Space Construction Framework for Multi-Task Learning**

**Chi-Yao Huang** et al. *ICLR 2026*. Proposed an orthogonal latent-space construction framework that decomposes task-specific factors into structured subspaces to reduce representation collapse. ([Web](#) | [Paper](#) | [Code](#) | [Talk](#))

**VOCAL: Visual Odometry via ContrActive Learning**

**Chi-Yao Huang** et al. *WACV 2026*. Proposed contrastive representation learning that organizes latent representations according to camera motion, producing interpretable motion-aware structure for visual odometry. ([Web](#) | [Paper](#) | [Talk](#) | [Demo](#))

## ONGOING RESEARCH

**Domain World Models: Layer-Structured Latent Representations for Cross-Domain Planning**

Developing a reward-free world model with layer-structured latent representations that enables agents to detect domain shifts, accurately predict future states, and plan across different environments.

**Temporal World Models: Dynamic Latent Representations for Time-Varying Environments**

Developing a temporal world model that explicitly captures time-varying factors in latent space, enabling agents to predict environmental changes and plan in dynamic environments.

**Object-Centric Learning with Contrastive Slot Attention**

Developing an object-centric representation-learning framework that integrates slot attention with contrastive learning to improve object discovery, slot consistency, and compositional visual reasoning.

INDUSTRY  
EXPERIENCE

**Senior R&D Engineer, HTC VIVE**

2017 – 2021

*Advanced and Creative Team, New Taipei, Taiwan*

- Developed core VIO/SLAM tracking systems deployed across VIVE COSMOS, VIVE FOCUS 3, VIVE XR Elite, and VIVE FLOW.
- **VIVE COSMOS:** Architected a multi-camera tracking system for consumer VR, achieving sub-millimeter tracking precision in controlled evaluation; developed a high-performance “mothership” SLAM system. ([Product](#) | [Demo](#))
- **VIVE FOCUS 3:** Led prototyping for a flagship MR system and optimized a tightly coupled visual-inertial SLAM pipeline on resource-constrained embedded platforms, achieving a 4× runtime speed-up. ([Product](#))
- **VIVE XR Elite:** Engineered low-latency VIO and obstacle-avoidance systems for mixed-reality interaction; designed core data structures for real-time spatial computing. ([Product](#))
- **VIVE FLOW:** Designed inside-out tracking for limited-FoV sensors and developed a motion-tracking prototype integrating real-time hand tracking. ([Product](#))
- **Spatial Mapping:** Built voxel-based obstacle mapping and semantic mapping modules for VR safety and real-world interaction.

SELECTED  
PROJECTS

**Toyota Mothership – Toyota Research Institute of North America (*Patent Pending*):** Developed a UAV pose-estimation framework using infrared imagery and IMU sensor fusion for high-altitude drone perception under challenging lighting and weather conditions. The underlying technologies are the subject of patent-pending applications. ([Demo](#))

**3D Landmark Reconstruction for Robot Localization and Mapping:** Implemented a learning-based 3D reconstruction pipeline that jointly optimizes camera trajectories and object-level semantic poses for robot localization. ([Demo](#))

**3D Reconstruction and Path Planning with Signed Distance Functions:** Fused dense image information with sparse feature points for efficient SLAM; implemented voxel hashing for dense mapping and integrated reconstruction with robot path planning.

**Humanoid Robot Control:** Developed bipedal walking patterns on non-planar surfaces by optimizing zero-moment-point trajectories and implemented localized A\* planning for obstacle navigation. ([Demo](#))

TEACHING &  
RESEARCH  
MENTORING

**Graduate Research Mentoring**

Mentored graduate students throughout their thesis research, providing guidance on problem formulation, literature review, experimental design, implementation, evaluation, and technical writing.

- **Zeel Bhatt (*OpalAI*)**  
Thesis on contrastive representation learning for visual odometry and robotics.
- **Khoa (Albert) Vo (*Ph.D. student, Purdue University*)**  
Thesis on multimodal reasoning and representation learning for embodied AI.
- **Shlok Mohanty (*Graduate Student*)**  
Thesis on representation learning from imbalanced robotic task data.
- **Hoang Quy Nguyen (*Graduate Student*)**  
Thesis on object-centric representation learning for structured visual understanding and downstream reasoning.

**Teaching Assistant, CSE 571: Artificial Intelligence**

*Arizona State University*

Supported course instruction through office hours, assignment grading, technical guidance, and individualized student support.

HONORS,  
AWARDS  
& GRANTS

**Research Grant, Toyota Research Institute of North America**

2023 – Present

**Ira A. Fulton Schools of Engineering Fulton Fellows Award, Arizona State University**

**Student with Distinction, Arizona State University**